When Correctness Is the Prediction: An Evaluation Trap in Sampling-Based Uncertainty for Handwritten Mathematics

Abstract

Sampling a vision–language model repeatedly and measuring how much it disagrees with itself is a cheap, label-free uncertainty signal. We evaluate it on handwritten mathematics and report one positive result, one negative result, and one way the evaluation itself manufactures a positive result that is not there. On transcription the signal works: AUROC 0.835, replicated at 0.828 on a second, architecturally independent model family, robust in both cases to every artifact control we apply, better than the model's own token confidence by a resolved margin, and better than simpler reductions of the same samples; a rule that abstains whenever all five samples disagree flags a genuinely wrong transcription 57 times out of 61. On grading it does not work: AUROC 0.520 on a balanced sample, indistinguishable from chance. The trap lies between the two. Stratifying the grading task by its ground-truth label — a natural response to a model with a strong answer bias — appears to recover a strong effect (0.801–0.854) that replicates across three independently trained model families. It is an artifact. Within a single-label stratum, "was the model correct" is the same column as "what did the model answer", so an uncertainty measure computed from those same answers predicts correctness almost by construction; a signal-free biased coin attains a *higher* AUROC than any real model we tested. We report the result we initially believed, the check that retired it, and what distinguishes the transcription task, where the same measure is sound. **[TODO: trim to the venue's word limit once the body is final]**

1 Introduction

[TODO: write. Do not open with "Large language models have recently…"]

2 Related Work

[TODO: write. VERIFY EVERY CITATION AGAINST THE REAL PAPER – do not cite from memory. This section is the largest genuinely-new writing task.]

3 Setup

Data. FERMAT [1] contains ∼2,200 photographed handwritten solutions to middle- and high-school problems, in which errors were deliberately introduced by the dataset authors and annotated by type. Each image contains a complete question and its answer, and 15% of items are error-free. That last property is load-bearing for this paper: §4.8 shows the analysis in §4.5 is impossible on datasets that lack it. Unless stated otherwise we use a balanced $n = 300$ sample (150 with an error, 150 without), drawn once with a fixed seed.

Tasks. We evaluate two distinct tasks on the same images. *Perception* asks the model to "explicitly perform OCR on the handwritten text and extract the content in LaTeX format"; it is correct when the extracted final answer matches the reference after canonicalisation. *Reasoning* asks it to "analyze the Answer to determine whether there is any error" and emit a binary judgement; it is correct when that judgement matches the annotation. The model is never shown the reference answer.

Uncertainty measure. For each item we draw $K = 5$ samples at temperature 0.7, reduce each to a canonical label, and take the Shannon entropy of the resulting cluster distribution, $H = -\sum_i p_i \ln p_i$. $H = 0$ when all samples agree and $H = \ln K$ when all differ. We deliberately use exact-match clustering over an *extracted answer span* rather than semantic clustering of whole derivations; §6 records why.

Models. Qwen2.5-VL at 3B and 7B, Pixtral-12B, LLaVA-NeXT-7B and InternVL3-8B — four independently trained families, all run in bfloat16.

Statistics. Every AUROC carries a 95% bootstrap interval over $10{,}000$ item-level resamples. Comparisons between conditions use a *paired* bootstrap over the same resampled items rather than checking whether two marginal intervals overlap. Before any run we registered a minimum minority-class size of 30 and, for the stratified reasoning hypothesis, an AUROC threshold of 0.70; verdicts below are assigned by a pure function of those thresholds, not by inspection.

4 Results

4.1 Does self-disagreement predict transcription error?

Yes, and by a margin that does not depend on the model's own confidence being poor. On the balanced $n = 300$ sample, perception entropy predicts transcription error with AU-

Signal	AUROC (95% CI)	vs. entropy
Sampling entropy	0.835 [0.787, 0.879]	—
1−majority fraction	0.793 [0.742, 0.841]	+0.042 [+0.011, +0.073]
Distinct count	0.790 [0.739, 0.838]	+0.045 [+0.016, +0.075]
Majority margin	0.779 [0.727, 0.830]	+0.055 [+0.018, +0.092]
Token log-probability	0.537 [0.469, 0.605]	+0.297 [+0.214, +0.380]

Table 1: Perception baselines, Qwen2.5-VL-3B, $n = 300$. The upper block reduces the same five samples in different ways; the lower is a model-internal signal requiring no sampling. Differences are paired bootstraps over the same resampled items, and every one excludes zero. Entropy's margin over the simpler summaries is small but resolved, so the shape of the cluster distribution carries information that a count or a majority does not. The simple summaries are related to entropy but not redundant with it (Spearman 0.83–0.89).

Subset	n	AUROC (95% CI)
All items	300	0.835 [0.787, 0.879]
Excl. parse failures	255	0.839 [0.790, 0.886]
Excl. max-entropy items	239	0.794 [0.737, 0.848]
Excl. both	206	0.796 [0.736, 0.852]

Table 2: Perception entropy under artifact controls, Qwen2.5-VL-3B. The signal is graded rather than concentrated at the ceiling: it survives the harshest simultaneous cut with an interval excluding chance. §4.9 shows a model for which it does not.

ROC 0.835 [0.787, 0.879]. The model's own mean token log-probability reaches only 0.537 [0.469, 0.605] — an interval that includes chance — and the paired difference is +0.297 [+0.214, +0.380], resolved. The signal is therefore not recoverable from information the model already exposes.

Read as a deferral rule the result is directly usable: of the 61 items on which all five transcriptions disagree, 57 are wrong (93.4% precision, 31.3% recall). Over the full risk–coverage curve this gives AURC 0.410 against a 0.607 random-deferral baseline. **[TODO: Fig. 1: risk–coverage curve. Asset exists at report/figures/risk_coverage_n300.png]**

4.2 Does it replicate on a second model family?

Yes. Pixtral-12B, architecturally independent of Qwen, reaches 0.828 [0.782, 0.871] on the same 300 images at a comparable transcription accuracy (41.7% against 39.3%). The intervals overlap along almost their whole length, so the two are not resolvably different.

More important than the point estimate is that it survives the control of §4.4: 0.772 [0.710, 0.833] after removing both parse failures and maximum-entropy items, with 17% of items at the ceiling. §4.9 describes a fourth model that scores *higher* raw and does not survive that control, which is the reason we report the two together.

4.3 What does the entropy summary add?

All five samples per item support several summaries, and a natural objection is that entropy is merely a dressed-up count of distinct answers. Table 1 tests that directly. The three sample-based alternatives are computed from the *same* generations at no additional cost, so the comparison isolates the summary rather than the sampling budget.

Two of these deserve comment. The majority margin separates a three–two split from a three–one–one split, which the majority fraction cannot, and it is nonetheless the weakest of the three — the information entropy uses is in the full distribution, not in the top two clusters. Token log-probability is the only baseline here that is not a function of the samples, and it is also the only one that fails to beat chance; a model that is verbally or numerically confident about a page it has misread is not detectable from its own output distribution, but is detectable from asking it twice. **[TODO: we also have verbalized confidence, but only on the grading arm (0.547 [0.432, 0.661], no signal, n=100). Either run it for perception or report it in §4.5 where the data exists — do not quietly omit it.]**

4.4 Does the signal survive artifact controls?

A cluster-entropy AUROC can be manufactured two ways that have nothing to do with useful uncertainty. Unparseable samples receive their own cluster label, which both inflates entropy and correlates with being scored wrong, so the metric could be measuring the parser. And items pinned at the ceiling $H = \ln K$ can carry the entire ranking if that cell happens to be uniformly incorrect, leaving a binary breakdown detector rather than a graded signal. Table 2 removes each, and both.

4.5 Does the same signal work for grading?

No. On the same 300 images, reasoning entropy predicts grading error with AUROC 0.520 [0.458, 0.582] — an interval containing chance. The balanced sample matters: the model answers "there is an error" for the large majority of inputs, and on an unbalanced set that bias alone would score well.

This is the honest reasoning result, and §4.6 is an account of why we did not initially believe it.

4.6 The trap: stratifying by the label being predicted

A model with a strong answer bias makes a pooled statistic hard to read, and the natural response is to stratify by ground truth and analyse each label separately. Doing so appears to recover a large effect from the null above. Within the erroneous-solution stratum, entropy predicts grading error at 0.801 [0.751, 0.846]; within the correct-solution stratum it predicts in the opposite direction at 0.280 [0.200, 0.366]. Both strata are adequately powered, both intervals exclude chance, the sign reversal has an appealing mechanistic story, and the

Model	Observed	Bias-only null	Collapse
Qwen2.5-VL-3B	0.854	0.901 [0.849, 0.932]	99.8%
Qwen2.5-VL-7B	0.801	0.868 [0.821, 0.903]	100.0%
LLaVA-NeXT-7B	0.775	0.831 [0.770, 0.874]	97.2%

Table 3: The apparent replication, against a null with no item-level signal at all: a model answering "error" with fixed probability per sample, so its entropy is pure sampling noise. The null *exceeds* every observed value. "Collapse" is the fraction of items on which correctness and the model's own verdict are the same label. Even the null's 2.5th percentile clears the 0.70 threshold we had registered in advance as confirming the hypothesis, so that threshold could never have separated signal from arithmetic.

effect reproduces across three independently trained model families (Table 3). We reported this as our main finding.

It is an artifact, for a reason that applies to any binary-decision task evaluated this way. **Within a stratum where every item shares the same true label, "was the model correct" is not a separate fact from "what did the model answer" — the two are the same column.** On our 7B run they agree on 100% of items, and the AUROC against correctness is numerically identical to the AUROC against the model's own verdict. Entropy is computed from precisely the votes that produce that verdict, so the measurement is near-circular. The sign reversal follows by the same identity: in the correct-solution stratum correctness is the exact *complement* of the verdict, so anything correlating with the verdict must invert. The two stratum AUROCs sum to 1.

The mechanism is arithmetic rather than behavioural. On an all-erroneous stratum the majority vote is wrong only when most samples dissent from the model's default, and dissent is exactly what raises entropy. Nothing about the model's competence is required for the association to appear.

Three things would have caught this earlier, and none of them is specialised. Comparing against a signal-free null rather than against chance, since 0.5 is the wrong reference point for a stratified statistic. Checking whether the correctness label is a relabelling of the prediction before computing anything. And treating a pre-registered threshold as necessary but not sufficient — ours was cleared by the null.

4.7 Why transcription is not affected

The collapse requires correctness to be a function of the vote *count*, which is also what entropy summarises. Transcription correctness instead depends on *which* label wins, so agreement does not imply correctness: 38 items are unanimous and 3 of those are wrong, while 4 items at maximum entropy are right. Correct and incorrect items occur at both ends of the entropy range, the label is not a relabelling of the prediction, and §4.3 shows entropy outperforming simpler summaries of the same votes — which could not happen if it were merely counting dissent.

4.8 What happens on a second dataset?

The analysis in §4.5 requires a dataset containing correct solutions, and the two public alternatives we examined do not have any: in ErrorRadar [2] no item is annotated error-free, and in ScratchMath [3] the student answer never equals the reference. Neither can be balanced, and neither can exhibit the inversion.

On ScratchMath, which is the only one of the two with handwriting to grade, the measurement fails for a second and more instructive reason. Grading accuracy reads 96.0%, but every item contains an error and the model answers "error" for 90% of them, so it scores well by construction. In 24% of readings (120/500) the model explicitly states it cannot read the image, and in 82% of those (98/120) it answers "error" regardless. Excluding those readings does not clean the estimate: all four misgraded items lie among them, so the retained subset has no minority class and the AUROC is undefined rather than improved. We therefore report this setting as *gated*: not a negative result about uncertainty, but a statement that the pair cannot test it.

4.9 When does a high score mean nothing?

InternVL3-8B attains the highest raw perception AUROC in our study, 0.915 [0.880, 0.944], on a transcription accuracy of 21.0% — about half that of the model in §4.1. Independently recomputing every derived column from raw output reproduces the stored values exactly, so this is model behaviour rather than a scoring defect. Under the control of §4.4 it collapses to 0.556 [0.495, 0.614].

The mechanism is visible in the entropy distribution: 202 of 300 items sit at the ceiling, and accuracy within that group is 0.99% against 62.2% elsewhere. The score is separating total generation breakdown from everything else, which is a binary property of the output, not graded uncertainty about content. Reported without the control, it would have been the strongest number in this paper. **[TODO: Fig. 2: the four-quadrant figure (low/high entropy × correct/wrong) with real pages. Assets exist at reference/cases/01–04.]**

5 Failure Analysis

[TODO: expand to the 50–100 coded cases. Tooling exists (pilot/07_manual_case_inspection.ipynb); this is reading time, not compute. Report inter-rater agreement if a second reader is available.]

6 Limitations

The positive result rests on one dataset. It now holds on two model families (§4.2), but two further models could not test it: one sits at a transcription floor of 0.8%, and one fails the artifact control of §4.4. Two of four attempts producing a usable measurement is itself worth stating — the result requires a model with genuine dense-OCR competence, and that is not the common case among open-weight VLMs. Of the two alternative datasets, neither contains correct solutions, so neither can even be balanced — and, by §4.6, an all-erroneous dataset is a single-label stratum, on which the measurement degenerates for the same reason.

We report the retraction of §4.6 as a finding, but it was found late. It was not caught by the pre-registration, which set a threshold the null clears; nor by adequate power, which both strata had; nor by replication, which reproduced the artifact three times. Cross-model agreement is worth less than it appears when the mechanism is arithmetic rather than behavioural.

Our clustering is exact-match over an extracted answer span, not semantic. Semantic clustering was attempted and abandoned: an NLI-plus-union-find implementation merges opposed clusters through a single false-positive transitive link and degrades at larger K. The reported results avoid that path rather than depending on it being fixed.

Confidence intervals were added after the experiments were designed and run. They corrected several conclusions but cannot supply power the studies were not sized for. $K = 5$ also gives few operating points, which is why two-thirds of conformal calibration splits cannot reach a 30% risk target.

7 Conclusion

On handwritten transcription, self-disagreement across repeated samples is a usable error signal: it survives every control we apply, beats the model's own confidence and simpler summaries of the same samples, and yields a deferral rule that is right 57 times in 61. On grading, the same measure carries no signal at all.

The distance between those two sentences is where the work is. Stratifying the grading task by ground truth — an ordinary response to a biased model — produces an effect of 0.80–0.85 that is adequately powered, resolves against chance, clears a pre-registered threshold, and replicates across three independently trained model families. It is still an artifact, because within a single-label stratum the correctness label is the model's own answer under another name. A signal-free biased coin scores higher.

We therefore recommend, for any evaluation of sampling-based uncertainty on a binary decision task: check whether the correctness label is a relabelling of the prediction before computing anything, and compare against a signal-free null rather than against 0.5. Neither is expensive. Both would have saved us a result we believed for three weeks.

References

[1] "Can vision-language models evaluate handwritten math?," *arXiv preprint arXiv:2501.07244*, 2025. VERIFY authors and venue.

[2] "Errorradar: Benchmarking complex mathematical reasoning of multimodal large language models via error detection," *arXiv preprint arXiv:2410.04509*, 2024. VERIFY authors and venue.

[3] "Scratchmath." VERIFY – dataset used via HuggingFace songdj/ScratchMath; find and cite the accompanying paper.